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MUBADALA

DxVar

AI-Driven Genomic Variant Analysis Tool

Collaborative Student Research Project

Final Report

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Abstract

Genomic variant interpretation is essential for understanding genetic disorders but is often hindered by complexity and resource demands. To address these challenges, we developed DxVar, an AI-driven variant ranking pipeline that leverages large language models to automate and accelerate variant interpretation and identify variant - phenotype associations. The developed pipeline consists of three stages. (1) extraction of Human Phenotype Ontology (HPO) terms from unstructured clinical descriptions using models such as ChatGPT-4, DeepSeek, LLaMA, and Doc2HPO; (2) variant prioritization using Exomiser based on pathogenicity scores, allele frequency, and phenotype-genotype associations; and (3) AI-assisted re-ranking of candidate variants using biomedical databases and literature to refine prioritization and generate interpretive summaries. The system is delivered through an intuitive Streamlit interface, offering variant annotation, linked phenotype literatures, ClinGen disease links, and AI summaries explaining the reasoning behind the ranking. Benchmarking on 10 GeneBreaker rare disease cases demonstrates that while Exomiser accurately identifies the causative gene in most instances, DxVar's AI module can further refine the ranking, placing the correct variant first in all analyzable cases. DxVar represents a step toward faster, more interpretable genomic analysis, with potential for further refinement through expanded datasets and improved phenotype modeling.

Introduction

Overview of the Project

The pipeline starts by extracting HPO terms from unstructured clinical text using models like LLaMA, ChatGPT-4, DeepSeek, and Doc2HPO. Alongside this, it automatically generates a JSON configuration file that includes case-specific HPO terms and tailored inheritance mode frequencies, ensuring relevant variants are not filtered out prematurely.

These inputs are passed to Exomiser, which performs variant annotation and prioritization based on allele frequency, pathogenicity scores, and phenotype-gene associations. The top-ranked genes and variants are then forwarded to DxVar's final AI module for interpretation and summary.

The entire process is accessible through a Streamlit interface, providing a fast, interpretable, and clinically relevant variant prioritization tool.

Primary Objectives

The goal of DxVar is to streamline and accelerate and improve the accuracy of genetic variant interpretation by:

1. Automating Genetic Variant Classification Across Formats

- Enable input flexibility by supporting **multiple variant formats** including rsIDs, chromosomal coordinates, and positional descriptions.
- Automatically interpret **rsIDs** using dbSNP to retrieve alleles and genomic context.
- Integrate with **GeneBe** and **InterVar** APIs to fetch **ACMG classifications** and variant effect predictions.

2. Leveraging AI for Disease Explanation and Variant Insight

- Use **LLaMA-3.3-70b** to provide detailed, AI-generated explanations of **Mendelian diseases**, helping users understand clinical implications of variants.
- Allow users to explore **variant-disease relationships interactively** through a chat interface that understands natural language.
- Ensure outputs are **structured and explainable**, supporting confident decision-making in both diagnostics and discovery.

3. Extract and Prioritize Using Phenotypic Data

- Automatically extract **HPO terms** from clinical descriptions using LLMs like **ChatGPT**, **LLaMA**, **DeepSeek**, and **Doc2HPO**.
- Integrates with **Exomiser** to prioritize variants based on **phenotype-gene associations**, **allele frequency**, and **predicted pathogenicity**.

4. Enable AI-Guided Re-Ranking of Top Variants

- Re-rank top variant candidates from Exomiser using **ACMG classifications** (GeneBe, InterVar), **PubMed-based literature mining** (PaperScraper), and **ClinGen and NCBI eUtils** for gene-disease scores.
- Provide an intuitive dropdown and summary system to aid **comparative evaluation** of variant relevance.

5. Create a Scalable and Expandable Platform

- Design DxVar to be modular and **easily upgradable**, supporting new models, databases, or languages.
- Enable long-term application in **clinical diagnostics, academic research, and precision medicine pipelines**.

Methodology

Pipeline Flowchart

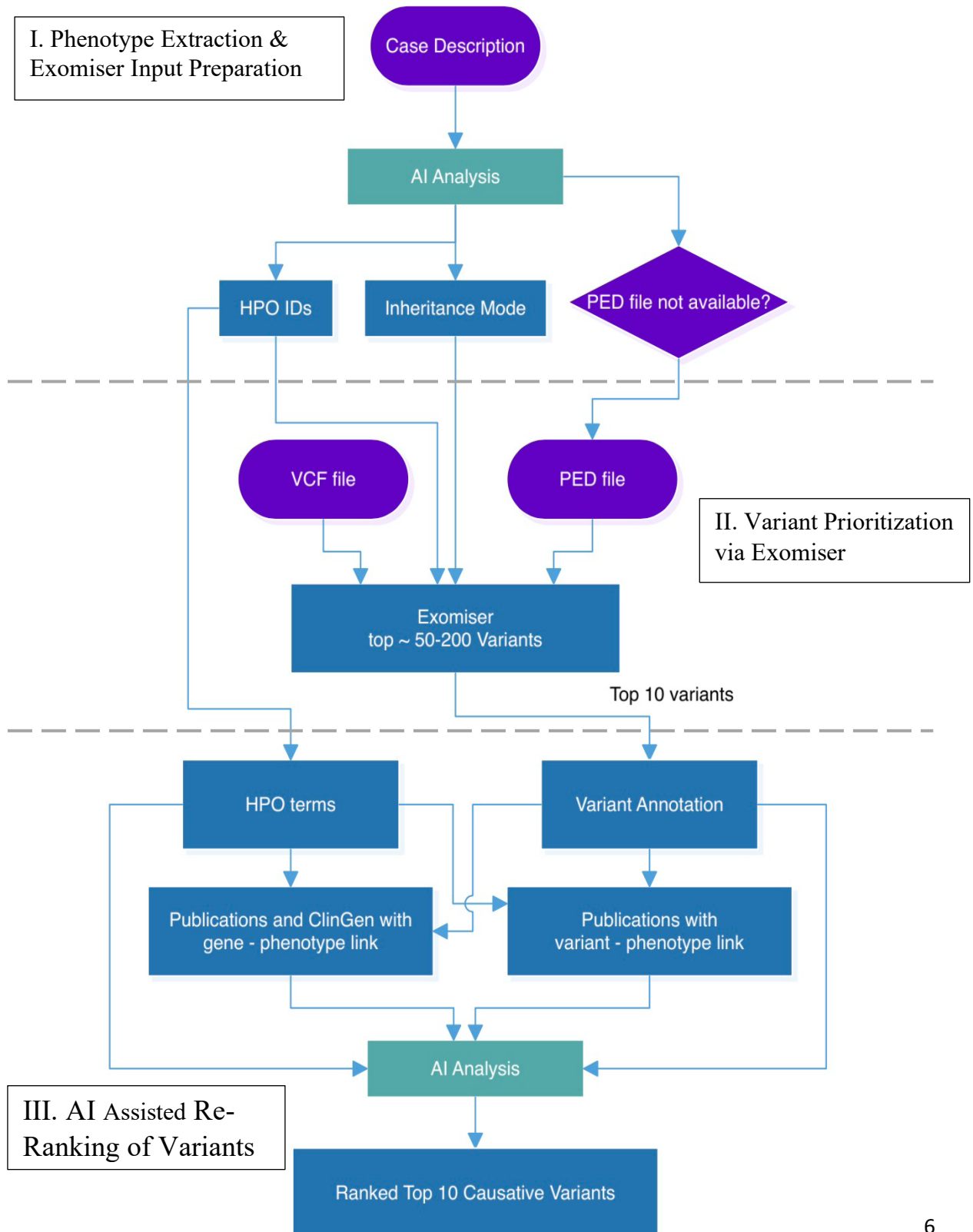


Figure 1: DxVar pipeline flowchart

I. Phenotype Extraction & Exomiser Input Preparation

The first step in the DxVar pipeline focuses on extracting clinically relevant phenotypic information from rare disease case descriptions. We begin by processing the raw clinical narrative using various AI models to extract corresponding **Human Phenotype Ontology (HPO) terms**. These terms encode patient symptoms and clinical features in a standardized format, which is essential for downstream variant prioritization.

In addition to extracting HPO terms, some of the AI models—specifically **ChatGPT-4, DeepSeek, and LLaMA 3.3**—are also leveraged to infer **likely inheritance modes**, including **Autosomal Dominant, Autosomal Recessive (Compound Heterozygous and Homozygous Alternate), X-linked Dominant, X-linked Recessive (Compound Heterozygous and Homozygous Alternate)**, and **Mitochondrial**. These inferences are based on clinical cues such as sex, consanguinity, family history, and disease onset. Notably, **Doc2HPO** does not provide inheritance prediction and is used only for HPO term extraction.

The extracted terms and inferred modes are structured into a JSON format compatible with Exomiser, enabling standardized and reproducible input. This preparation step ensures high compatibility with downstream tools and enhances the clinical utility of phenotype-driven variant filtering.

To assess the effectiveness of this process, we evaluated DxVar’s extraction and formatting pipeline on 10 benchmark clinical cases from **GeneBreaker**. Each tool was tested on the same input cases, and the extracted outputs were compared against GeneBreaker’s reference HPO annotations. This comparison allowed us to measure the reliability and performance of each model in real-world diagnostic scenarios.

In summary, this step of the DxVar pipeline transforms unstructured clinical narratives into structured phenotype data and predicted inheritance modes. These components serve as the foundation for variant interpretation using Exomiser. Our evaluation revealed both the strengths and limitations of each tool, guiding future refinement in phenotype extraction and inheritance prediction for rare disease diagnostics.

II. Variant Prioritization via Exomiser

The DxVar pipeline incorporates Exomiser as its core variant prioritization module, applying it to structured clinical data (HPO terms) and genomic variant files (VCFs) to systematically identify and rank candidate disease-causing variants. This scoring framework enables DxVar to output a focused, interpretable, and actionable list of variants aligned with the clinical presentation.

Exomiser Integration

Exomiser is embedded within the DxVar system to function as the variant filtering and ranking engine. Rather than using the tool in isolation, it is dynamically configured and executed based on upstream outputs and case-specific parameters produced by the DxVar pipeline.

The integration involves the following coordinated steps:

Input Data Preparation

- VCF files from the user are aligned to the correct genome assembly (GRCh37 or GRCh38) to ensure compatibility with Exomiser's backend.
- HPO terms are extracted automatically through the upstream AI modules in DxVar (e.g., LLaMA, DeepSeek), forming the key phenotypic input to guide prioritization.
- Inheritance-relevant information is inferred from the case description and translated into a .ped file or JSON structure to inform the prioritization logic.

Inheritance Mode Determination

- DxVar assigns appropriate inheritance modes using language-based inference from the clinical description, enabling flexible prioritization in uncertain or atypical genetic scenarios.
- Inheritance mode thresholds within Exomiser's configuration are automatically tuned for each case, ensuring important variants are not prematurely excluded.

Frequency-based Filtering

- Variants are automatically filtered according to population allele frequencies pulled from integrated databases like gnomAD, 1000 Genomes, and others.
- The pipeline relaxes thresholds during early filtering passes to preserve rare but potentially relevant variants, especially when case-specific genetic models are ambiguous.

Pathogenicity Scoring and Variant Annotation

- DxVar leverages Exomiser's built-in annotation capabilities to enrich each variant with pathogenicity predictions sourced from tools like REVEL, MVP, and ALPHA_MISSENSE.

- These scores are integrated into the final ranking to highlight clinically significant variants while retaining broader candidate coverage.

Phenotype-Gene Association

- Extracted HPO terms are compared to known gene-phenotype associations using data from ClinGen and OMIM.
- DxVar uses this match to elevate genes whose functions align closely with the described clinical features, further refining the prioritization results.

Ranking Mechanism

DxVar utilizes Exomiser's multi-layered scoring engine, which combines:

- Inheritance model fit derived from inferred family structures.
- Population frequency checks to exclude common polymorphisms.
- Phenotypic relevance via HPO-term alignment.
- Pathogenicity assessment from integrated prediction scores.

The top 10 genes produced by Exomiser are then passed forward within the DxVar pipeline for final AI-powered interpretation. This final stage generates a structured clinical summary for each gene and its associated variants, helping users understand the rationale behind each prioritization and how it links to the case description.

III. AI Assisted Re-Ranking of Variants

After retrieving the ranked variants from Exomiser's filtering, the HPO IDs and top 10 variants are run through an AI assisted re-ranking to find the variants most likely to cause the observed phenotypes. This step uses multiple API's and biomedical databases to gather relevant information for the AI to make informed ranking decisions. This information includes the following:

- Annotated variants via Genebe and Intervar providing classification and detailed variant information
- Variant – Phenotype literatures via paperscraper using PubMed database
- Gene – Phenotype relationship using literature counts with NBI's eutils API using the PubMed database and with ClinGen's database to get established Gene – Disease links

An AI overall summary is then created using all the compiled data. Individual variants can be selected from a dropdown list to display their details and relevant information. The workflow is illustrated in the flowchart below.

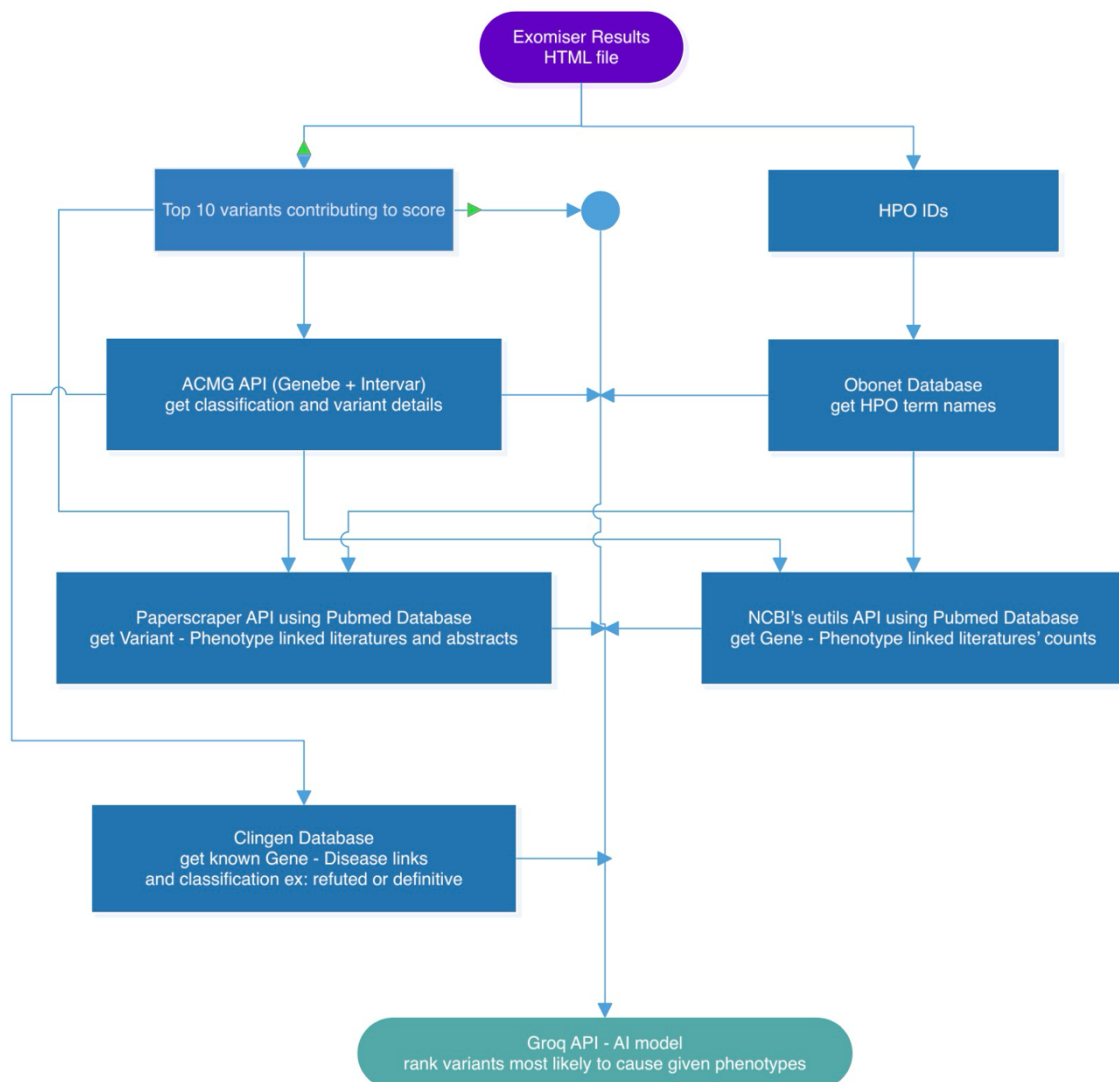


Figure 2: DxVar(III) flowchart

Interface



Upload Exomiser HTML file

 Drag and drop file here
Limit 200MB per file • VCF, TXT, JSON, HTML, HTM

Browse files

 result2.5.html 0.9MB

×

Select variant to view:

Variant 1: 2,47403219,C,T,hg38

Generate Overall AI Summary

ACMG Results: **Pathogenic**

⬇ 🔍 ↺

Attribute	GeneBe Results	InterVar Results
Classification	Pathogenic	Pathogenic
Effect	stop_gained	
Gene	MSH2	MSH2
HGNC ID	7325	
dbsnp	rs63751099	
freq. ref. pop.	None	
acmg score	18	
acmg criteria	PVS1,PM2,PP5_Very_Strong	

ClinGen Gene-Disease Results

DISEASE LABEL	MOI	CLASSIFICATION	DISEASE ID (MONDO)
Lynch syndrome	AD	Definitive	MONDO:0005835
mismatch repair cancer syndrome 1	AR	Definitive	MONDO:0010159
hereditary breast carcinoma	AD	Refuted	MONDO:0016419

Research Papers

6 Research papers were found related to this variant.

Abnormal eye morphology (0 papers)

Episodic vomiting (0 papers)

Abdominal pain (0 papers)

Maternal teratogenic exposure (0 papers)

Macrovesicular hepatic steatosis (0 papers)

Branchial cyst (0 papers)

Decreased platelet glycoprotein IIb-IIIa (0 papers)

Abnormal fifth cranial nerve physiology (0 papers)

Neoplasm (4 papers)

	title	journal
1	Whole-exome Sequencing of Nigerian Prostate Tumors from the Prostate Cancer Tran	Cancer research commun
2	Prevalence and spectrum of DNA mismatch repair gene variation in the general Chine	Journal of medical genet
3	Leveraging a Multi-Omics Strategy for Prioritizing Personalized Candidate Mutation-D	Scientific reports
4	Data integration workflow for search of disease driving genes and genetic variants.	PloS one

Short stature (0 papers)

Clinical course (0 papers)

Generate AI Summary for this variant

Figure 3: DxVar(III) Streamlit hosted app interface screenshot showing ability to select variant to view its related info including ACMG results, ClinGen Gene-Disease links and literature papers with variant - phenotype link

The figures above show the interface developed for the final step in the pipeline. It is hosted on Streamlit at <https://dxvarchat.streamlit.app/>. With the code being publicly available on the following GitHub repository: <https://github.com/DxVarTeam/DxVar>.

Results

The pipeline was tested across the 10 cases provided by GeneBreaker. The results are given below.

I. Phenotype Extraction & Exomiser Input Preparation

To evaluate the performance of our phenotype extraction pipeline, we benchmarked four different models and tools—**DeepSeek**, **ChatGPT-4**, **LLaMA 3.3**, and **Doc2HPO**—against 10 clinical cases provided by GeneBreaker. Each tool was assessed based on how accurately it could extract relevant Human Phenotype Ontology (HPO) terms and inheritance patterns. The key performance metric we used was **Matching Accuracy**, representing the proportion of correctly extracted terms compared to the reference.

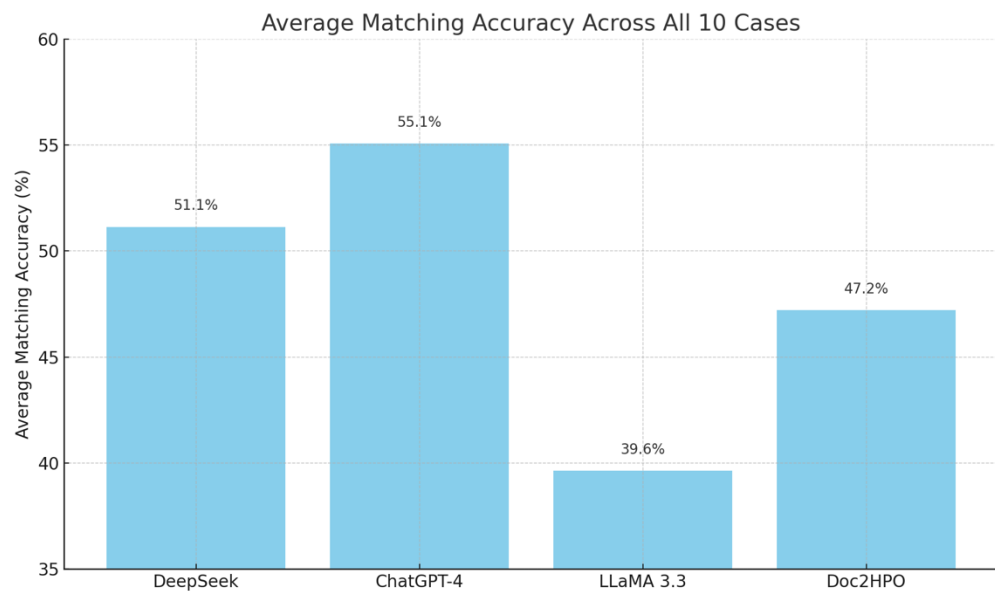


Figure 4: Accuracy of 3 AI models and the Doc2HPO tool in extracting HPO terms from given case descriptions in GeneBreaker benchmarked against the provided HPO terms.

The bar chart above visualizes the **average matching accuracy** across all 10 cases for each model. **ChatGPT-4** achieved the highest accuracy at **55.1%**, followed by **DeepSeek** at **51.1%**, while **Doc2HPO** and **LLaMA 3.3** trailed behind with **47.2%** and **39.6%** respectively. Although none of the tools consistently outperformed others across all cases, this comparison highlights ChatGPT-4 and DeepSeek as the most reliable options for phenotype extraction tasks in this dataset. Notably, all models struggled with correctly predicting inheritance, which remains an area for future improvement.

(II) Variant Prioritization via Exomiser & (III) AI Assisted Re-Ranking of Variants

Exomiser successfully ranked the causative gene first in seven out of ten cases. The gene was ranked 3rd and 4th in two cases, while in one instance it was completely filtered out. Using the top 10 variants identified by Exomiser, we applied AI-based re-ranking with both Llama and DeepSeek, which consistently ranked the correct variant first in all applicable cases. This outcome underscores the strength of the AI re-ranking approach but also highlights the importance of setting the appropriate parameters of Exomiser. The results are summarized in the table below.

Case	Causative Gene	Gene Rank		
		Exomiser (pipeline step II)	DxVar llama-3.3-70b- versatile	DxVar deepseek-r1- distill-llama-70b
1	WAS	1	1	1
2	MSH2	1	1	1
3	MALT1	4	1	1
4	CFTR	3	1	1
5	MECP2	1	1	1
6	ABCD1	1	1	1
7	EP300	1	1	1
8	PTPN11	Filtered Out	-	-
9	COL2A1	1	1	1
10	ANKRD11	1	1	1

Table 1: comparing the gene rankings of stage II and III of DxVar's pipeline as well as using 2 different AI models in stage III

Limitations

1. Dependence on Upstream Components

- The accuracy of variant prioritization depends on the quality of extracted HPO terms and inheritance predictions. Errors or omissions in earlier stages can cause Exomiser to miss or deprioritize the causative variant.
- Inheritance mode inference from case descriptions remains challenging, particularly in subtle or atypical scenarios.

2. Filtering Sensitivity in Exomiser

- Hard filtering thresholds for frequency and pathogenicity may exclude borderline or rare pathogenic variants.
- Once a variant is filtered out, it is not accessible to downstream interpretation, limiting opportunities for review or rescue.

3. External Database Dependency

- Exomiser's performance relies on the completeness of databases like OMIM, ClinGen, and gnomAD. Poorly annotated or novel genes may be missed or undervalued.

4. Limited Dataset Scope

- Testing was conducted on 10 synthetic GeneBreaker cases. While controlled and informative, this dataset does not reflect the diversity and complexity of real-world clinical data.

5. Re-ranking Model Constraints

- The AI re-ranking stage only evaluates what Exomiser outputs—missed variants cannot be recovered.
- AI outputs can vary between runs due to non-deterministic behavior, especially among lower-ranked variants.

Future Improvements

1. Enhanced Variant Prioritization

- Integrate additional prioritization tools (e.g., LIRICAL, AMELIE) alongside Exomiser. Use consensus-based overlap scoring to improve recall and confidence in top-ranked genes.

2. Smarter Filtering

- Replace static frequency and pathogenicity cutoffs with context-aware ML models that adjust thresholds based on phenotype severity, inheritance uncertainty, and rarity of the case.

3. Surface Excluded Variants

- Allow clinicians or AI models to review variants filtered out by Exomiser, enabling potential manual validation or recovery of missed findings.

4. Real-Time Feedback Integration

- Develop a feedback loop between the AI interpretation module and Exomiser, allowing downstream results to inform reconfiguration of thresholds for re-analysis.

5. Expand Knowledge Sources

- Enhance phenotype-gene mapping using real-time literature mining and functional genomic resources to improve support for under-annotated or novel genes.

6. Improved Evaluation and Dataset Diversity

- Validate the pipeline using real-world clinical datasets to assess generalizability, robustness, and performance across diverse case types.

7. Transparency and Debugging Tools

- Add a visual debugging interface that traces filtering steps and ranking decisions per case, increasing interpretability for clinicians.

Discussion

Paper Title	Publication year	Goal of the Study	Input Types	AI/LLM Model Used	Databases Used	Tools/Apps Used	Study Conclusion
1. Assessing the utility of large language models for phenotype-driven gene prioritization in the diagnosis of rare genetic disease	2024	- To evaluate five LLMs for phenotype-driven gene prioritization in rare disease diagnosis	- phenotype descriptions provided as structured HPO terms - free-text phenotypic descriptions from literature	- GPT-3.5 - GPT-4 - Llama2-7b-chat - Llama2-13b-chat - Llama2-70b-chat	- The study utilized publicly accessible datasets pooled from five sources (AJHG, CSH, ColumbiaU, DGD, and TAF1)	- bioinformatics tools (such as Phenomizer, Exomiser, AMELIE, Phenolyzer, Phen2Gene, DeepSVP, CADA, and GADO) - APIs: OpenAI API, Replicate's API, and the LlamaIndex library	- GPT-4 achieved the highest average overall prediction accuracy - LLMs exhibit a bias towards predicting highly cited genes and lag behind traditional tools in terms of prediction accuracy
2. Survey and Improvement Strategies for Gene Prioritization with Large Language Models	-	- To evaluate and enhance the use of LLMs for identifying causal genes in rare genetic diseases	- patient clinical phenotypes - lists of possible genes from genetic tests	- GPT-4 - GPT-3.5 - Mixtral-8x7B - Llama-2-70B, - BioMistral	- Patient cases provided by: Baylor Genetics (BG), the Undiagnosed Diseases Network (UDN), and Deciphering Developmental Disorders (DDD) - ClinVar and HGMD	- SpliceAI (variant scoring) - The get_ontology() function to retrieve the HPO data - HPO Phenotype Classification methodology, using the dataset specificity index (Dsl).	- GPT-4 consistently performed best in ranking causal genes - Multi-agent and HPO classification accounted for case differences, while the divide-and-conquer strategy mitigated biases and improved accuracy
3. Harnessing large language models (LLMs) for candidate gene prioritization and selection	2023	- To assess the utility of LLMs for knowledge-driven gene prioritization and selection to accelerate biomarker panel development	- transcriptome profiling data (NCBI GEO repository)	- GPT-3.5 - GPT-4 - Bard (Google) - Claude (Anthropic)	- NCBI GEO repository - PubMed	- BloodGen3 (module repertoire construction) - Gene eXpression Browser (GXB) - Ingenuity Pathway Analysis (IPA). - DAVID (functional annotation clustering)	- GPT-4 and Claude were identified as the most proficient models, chosen for workflow implementation after benchmarking - large input limits (seen in GPT-4) were overcome by using models with larger context windows (like Claude)
4. The application of Large Language Models to the phenotype-based prioritization of causative genes in rare disease patients	2025	- To evaluate the effectiveness of LLMs in prioritizing disease-associated genes based on phenotypes	- A list of candidate genes - Patient phenotypes: Natural language terms, HPO terms or codes, and observed phenotypes.	- GPT-3.5-turbo - GPT-4 - Falcon180B	- GPCards - ClinVar - Phenotype-Associated Variants in Saudi Arabia (PAVS) database - HPO database - OMIM - gnomAD	- Exomiser (version 12.1.0) - bcbio-nextgen toolkit - GATK Haplotype caller - Slivar (variant filtering)	- GPT-4 showed higher accuracy than Exomiser in ranking plausible and causative genes - LLMs, such as GPT-4, are robust against noise in patient phenotypes, tolerating alterations in phenotypes
5. Genetic Transformer: An Innovative Large Language Model Driven Approach for Rapid and Accurate Identification of Causative Variants in Rare Genetic Diseases	-	-To develop and evaluate Genetic Transformer (GeneT), an innovative LLM driven approach, for the rapid and accurate identification of causative variants	- Basic features: gene symbol, transcript, zygosity, and HGVS name. - ClinVar - population data. - ACMG info - Patient phenotypes (diseases, free-text, and HPO terms)	-Qwen1.5-0.5B -Qwen1.5-1.8B -Qwen1.5-4B -Mistral-7B -Meta-Llama-3-8B -Meta-Llama-3-70B	- HGMD (Human Gene Mutation Database) - OMIM - GPCards - 1000 Genomes Project	- BCFanno and SIGVAR (variant annotation) - PHIVE, PhenIX, Exomiser, and Xrare - PhenoTagger (extracts HPO terms from case description)	- the fine-tuned Qwen1.5-1.8B exhibited the most notable reduction in the number of candidate variants in synthetic datasets and achieved high recall rates - GeneT achieved a high recall rate >97% in both synthetic datasets and clinical settings and 20-fold increase in processing speed.

6. Enhancing phenotype recognition in clinical notes using large language models: PhenoBCBERT and PhenoGPT	2024	- To automate the detection of phenotype terms, including those not in the current Human Phenotype Ontology (HPO) by developing two large language models, PhenoBCBERT and PhenoGP.	- clinical abstracts from rare disease patients at Children's Hospital of Philadelphia (CHOP) - BiolarkGSC+ corpus - ID-68 dataset	- PhenoBCB ERT - PhenoGP - Open-source models: GPT-J-6B, Falcon-7B, and LLaMA-7B - Closed-source models: GPT-3 and GPT-3.5 were used. GPT-3 (davinci), and GPT-3.5 (gpt-3.5-turbo).	- Clinical notes from Children's Hospital of Philadelphia (CHOP) - Public datasets: BiolarkGSC+ and ID-68. - HPO database - Unified Medical Language System (UMLS) - PubMed - MIMIC III corpora - Orphanet Database	- PhenoTagger - MetaMap - NCBO annotator - Doc2HPO - ClinPhen - Neural Concept Recognizer (NCR) - PhenoBERT - PhenoRerank - CLAMP - PyTorch module and HuggingFace interface	- Both PhenoBCBERT and PhenoGPT models demonstrated the capability to extract phenotypic information with comparably high precision, recall, and accuracy on public datasets - PhenoBCBERT was found to be more robust to typos and misspellings than PhenoTagger, and both PhenoBCBERT and PhenoGPT could recognize misspellings present in published data
7. Improving Automated Deep Phenotyping Through Large Language Models Using Retrieval Augmented Generation	-	- To present RAG-HPO, a Python-based tool that leverages RAG to elevate LLM accuracy HPO term assignment	- free-text provided as a string, either supplied directly by the user or batch-processed from a CSV file containing clinical notes	- Llama-3 70B model	- HPO database	- RAG-HPO - Doc2HPO - ClinPhen - FastHPOCR - Facebook AI Similarity Search (FAISS)	- RAG-HPO demonstrated superior precision and recall compared to traditional HPO tools - RAG-HPO minimizes the need for fine-tuning and reduces the risks of "hallucinations," unlike conventional LLM tools.
8. A simplified retriever to improve accuracy of phenotype normalizations by large language models	2025	- To improve phenotype normalization accuracy using simplified BioBERT retriever combined with LLMs	- Phenotypic terms derived from OMIM	- GPT-4o - GPT-3.5-Turbo - GPT-4o-mini	- HPO Database - OMIM - NCBO BioPortal	- BioBERT v1.1 model - spaCy - Doc2Hpo	- GPT-4o with retriever augmentation showed a significant increase in accuracy from a baseline of 62% to 85%. - GPT-4o and GPT-3.5-Turbo with the retriever achieved the highest accuracy of 85% to 88%.

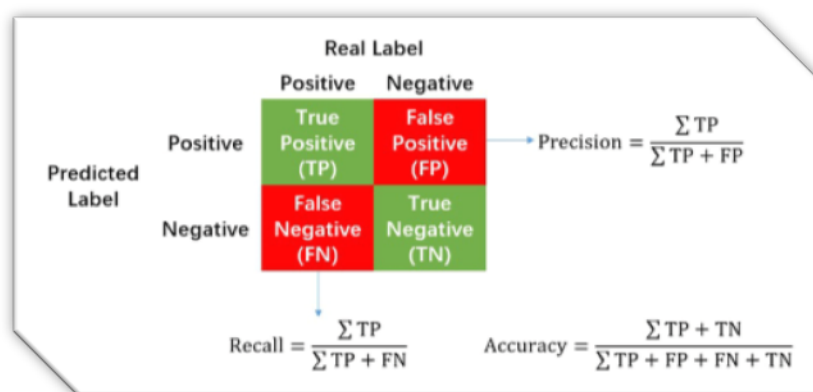
Table 2: Summary of Recent Studies Applying Large Language Models (LLMs) in Genomic Variant Interpretation and Gene Prioritization

Appendix

Phenotype Extraction & Exomiser Input Preparation – Case Results

Case 1:

“A two-year-old boy presents to his pediatrician with bloody diarrhea, eczema, petechiae and easy bruising with no associated trauma. He has had multiple treatments for otitis media (ear infections). Investigations reveal thrombocytopenia (decreased number of platelets) and abnormal morphology (small platelets). The proband is an only child. Both parents do not have any significant medical problems. The family history is significant for a maternal uncle with autoimmune problems and a recent diagnosis of lymphoma”.



Term	GeneBreaker (Ref)	DeepSeek	ChatGPT-4	LLaMA 3.3	Doc2HPO
Eczema	HP:0000964	✓	✓	✓	✓
Thrombocytopenia	HP:0001873	✓	✓	✓	✓
Otitis media	HP:0000388	✗	✗	✗	✓
Hematemesis	HP:0002248	✗	✗	✗	✗
Inheritance	X-linked recessive	✓	✓	✓	✗
Matching Accuracy		50.0%	50.0%	50.0%	75.0%
Precision (%)		28.6%	28.6%	28.6%	37.5%
Recall (%)		50.0%	50.0%	50.0%	75.0%

In the table above, we compare the performance of four different models/tools—**DeepSeek**, **ChatGPT-4**, **LLaMA 3.3**, and **Doc2HPO**—against the reference results provided by **GeneBreaker**. The goal was to assess how accurately each model could extract the correct **Human Phenotype Ontology (HPO) terms** and **inheritance pattern** from a clinical case.

Each term listed (e.g., *Eczema*, *Thrombocytopenia*) is annotated with its corresponding HPO code from the GeneBreaker reference. A  indicates that the model/tool successfully extracted the correct term, while a  indicates a miss.

At the bottom of the table, we present three key metrics:

- **Matching Accuracy:** The percentage of exact matches with GeneBreaker’s output.
- **Precision (%):** The proportion of correctly predicted terms among all terms predicted by the model/tool.
- **Recall (%):** The proportion of correct terms identified out of all relevant terms in the GeneBreaker reference.

This evaluation helps identify which tool most closely aligns with expert-generated outputs, providing insight into each model’s clinical relevance and extraction accuracy.

Case 2:

“A 41-year-old male presents to his family doctor with fatigue and abdominal pain. Initial investigations included a complete blood count, renal and liver panels. He had normocytic anemia and mild elevations of hepatic enzymes. Imaging (CT) of the abdomen demonstrated several hypoattenuating lesions in the liver and edema of the bowel mucosa at the ileocecal junction. Colonoscopy revealed a large mass protruding into the lumen of the distal part of the ascending colon. Subsequent histopathology revealed an adenocarcinoma with negative margins. Microsatellite instability was high. There was no significant family history of cancer or other illnesses”.

Term	GeneBreaker (Ref)	DeepSeek	ChatGPT-4	LLaMA 3.3	Doc2HPO
Intestinal polyposis	HP:0200008	✗	✗	✗	✗
Seizure	HP:0001250	✗	✗	✗	✓
Hypertonia	HP:0001276	✗	✗	✓	✓
Fatigue	HP:0012378	✓	✓	✓	✓
Abdominal pain	HP:0002027	✓	✗	✓	✓
Inheritance	Autosomal dominant	✗	✓	✓	✗
Matching Accuracy		40.0%	40.0%	20.0%	40.0%
Precision (%)		25.0%	25.0%	12.5%	50.0%
Recall (%)		40.0%	40.0%	20.0%	40.0%

In this comparison for Case 2, we evaluated each model/tool—DeepSeek, ChatGPT-4, LLaMA 3.3, and Doc2HPO—against the GeneBreaker reference to determine how well they extracted relevant HPO terms and inheritance patterns from a clinical case. The metrics at the bottom (Matching Accuracy, Precision, and Recall) indicate how closely each model's outputs matched the expert-labeled terms. While all models showed limitations, **Doc2HPO** had the highest **precision** (50.0%) and tied for the best **matching accuracy** and **recall** (40.0%) with DeepSeek and ChatGPT-4.

Case 3:

“A 14-year-old female born to consanguineous parents (first cousins) with short stature and low weight presented with a history of the following: an eczematous rash since (onset when she was a neonate) and persistent severe dermatitis. She has had multiple skin infections and severe inflammatory gastrointestinal disease”.

Term	GeneBreaker (Ref)	DeepSeek	ChatGPT-4	LLaMA 3.3	Doc2HPO
Eczema	HP:0000964	✓	✓	✓	✓
Atopic dermatitis	HP:0001047	✗	✓	✗	✗
Recurrent skin infections	HP:0001581	✓	✓	✗	✗
Gastrointestinal inflammation	HP:0004386	✗	✗	✗	✗
Recurrent respiratory infections	HP:0002205	✗	✗	✗	✗
Pneumonia	HP:0002090	✗	✗	✗	✗
Inheritance	Autosomal recessive	✓	✓	✓	✓
Matching Accuracy		50.0%	50.0%	16.7%	16.7%
Precision (%)		50.0%	50.0%	16.7%	16.7%
Recall (%)		50.0%	50.0%	16.7%	16.7%

In Case 3, each model/tool was compared to GeneBreaker’s reference annotations to evaluate its ability to extract relevant HPO terms and inheritance pattern from a clinical scenario. DeepSeek and ChatGPT-4 demonstrated the best performance, each achieving 50% across all metrics—matching accuracy, precision, and recall. In contrast, LLaMA 3.3 and Doc2HPO detected fewer relevant terms, resulting in significantly lower scores (16.7%). This highlights clear performance variability across models when interpreting more complex or ambiguous phenotype cases.

Case 4:

“A two-year-old male presented to his pediatrician with a fever of unknown origin. He was underweight for height and had a history of loose stools. His work up demonstrated an elevated white count. His chest x-ray showed patchy atelectasis (partial collapse of the lung) and a R middle lobe infiltrate. He was admitted to the hospital for IV antibiotics and pseudomonas aeruginosa were isolated from culture from a sputum sample. He underwent sweat chloride testing which was positive. The family history revealed a healthy older sister. Parents were in good health and nonconsanguineous. His mother was of Scottish heritage, and his father was Irish”.

Term	GeneBreaker (Ref)	DeepSeek	ChatGPT-4	LLaMA 3.3	Doc2HPO
Biliary cirrhosis	HP:0002613	✗	✗	✗	✗
Immunodeficiency	HP:0002721	✗	✗	✗	✗
Malabsorption	HP:0000244	✗	✗	✗	✗
Pulmonary fibrosis	HP:0002206	✗	✓	✗	✗
Recurrent respiratory infections	HP:0002205	✗	✓	✓	✗
Exocrine pancreatic insufficiency	HP:0001738	✗	✗	✗	✗
Inheritance	Autosomal recessive	✓	✓	✓	✓
Matching Accuracy		0.0%	33.3%	16.7%	0.0%
Precision (%)		0.0%	22.2%	11.1%	0.0%
Recall (%)		0.0%	33.3%	16.7%	0.0%

In Case 4, we evaluated the ability of each tool—DeepSeek, ChatGPT-4, LLaMA 3.3, and Doc2HPO—to identify relevant HPO terms and inheritance from a complex case annotated by GeneBreaker. Most models struggled with this case, identifying very few correct terms. **ChatGPT-4** performed the best with a matching accuracy and recall of **33.3%**, while **LLaMA 3.3** followed at **16.7%**. DeepSeek and Doc2HPO failed to extract any correct terms, resulting in **0%** across all metrics.

Case 5:

“A two-year-old female presented for a well child check. Her early pediatric milestones were reassuring, she walked shortly after her first birthday and began talking around the same time. However, at the appointment, her mother notes that she is no longer using words that she used to and that she no longer feeds herself and seems less engaged. On examination, her growth parameters are in the normal range, but are notable for minimal growth in head circumference over the last 9 months. She verbalizes but does not make intelligible words and does not reciprocate in verbal exchange. She is an only child and the family history is negative with respect to significant medical problems”.

Term	GeneBreaker (Ref)	DeepSeek	ChatGPT-4	LLaMA 3.3	Doc2HPO
Seizure	HP:0001250	✗	✗	✗	✗
Spasticity	HP:0001257	✗	✗	✗	✗
Postnatal microcephaly	HP:0005484	✗	✓	✗	✗
Intellectual disability profound	HP:0002187	✗	✗	✗	✗
Inheritance	Autosomal recessive	✗	✗	✗	✗
Matching Accuracy		0.0%	25.0%	0.0%	0.0%
Precision (%)		0.0%	16.7%	0.0%	0.0%
Recall (%)		0.0%	25.0%	0.0%	0.0%

In Case 5, the tools struggled significantly to match GeneBreaker’s reference phenotype terms. Only **ChatGPT-4** successfully identified one correct term (*Postnatal microcephaly*), resulting in a **25.0% matching accuracy and recall**, and **16.7% precision**. All other tools—DeepSeek, LLaMA 3.3, and Doc2HPO—missed every term and inheritance label, scoring **0%** across all performance metrics. This case demonstrates the limitations of current tools in capturing complex or subtle clinical features.

Case 6:

“A six-year-old boy is seen by his pediatrician as his mother noted that he has become hyperactive, easily distracted and impulsive. His mother said he was behind with respect to language skills, fine motor skills and coordination. She also expressed concern to the physician that he continues to resist eating solid foods as he frequently chokes and gags when eating. He was a normal appearing male with a slight build. He had increased muscle tone with brisk reflexes at the patellae bilaterally. He was referred to neurology and brain imaging identified symmetric enhancement of signal in the parieto-occipital region with evidence of neuronal loss. The family history was otherwise negative for significant medical concerns. Parents are nonconsanguineous”.

Term	GeneBreaker (Ref)	DeepSeek	ChatGPT-4	LLaMA 3.3	Doc2HPO
Seizure	HP:0001250	✗	✗	✗	✗
Psychosis	HP:0002180	✗	✗	✗	✗
Primary adrenal insufficiency	HP:0000709	✗	✗	✗	✗
Neurodegeneration	HP:0008207	✗	✗	✗	✗
Abnormality of the cerebral white matter	HP:0002500	✗	✗	✗	✗
Inheritance	X-linked dominant	✗	✗	✗	✗
Matching Accuracy		0.0%	0.0%	0.0%	0.0%
Precision (%)		0.0%	0.0%	0.0%	0.0%
Recall (%)		0.0%	0.0%	0.0%	0.0%

Case 6 highlights a scenario where **none** of the evaluated models or tools—DeepSeek, ChatGPT-4, LLaMA 3.3, or Doc2HPO—were able to correctly identify **any** of the six HPO terms or the inheritance pattern provided by GeneBreaker. All performance metrics, including matching accuracy, precision, and recall, were **0.0%**, reflecting a complete failure in phenotype recognition for this complex case across all systems tested.

Case 7:

“A seven-year-old female was referred to Medical Genetics. She had moderate intellectual disability and characteristic craniofacial features that included: arched eyebrows, downslanting palpebral fissures, a prominent nose and micrognathia. She was also noted to have broad thumbs. The family history revealed that her father had significant learning problems (did not finish high school) and similar craniofacial features to his daughter”.

Term	GeneBreaker (Ref)	DeepSeek	ChatGPT-4	LLaMA 3.3	Doc2HPO
Moderate intellectual disability	HP:0002342	✓	✓	✓	✓
Arched eyebrows	HP:0002553	✓	✓	✗	✓
Downslanting palpebral fissures	HP:0000494	✓	✓	✓	✓
Prominent nose	HP:0000448	✓	✗	✓	✓
Micrognathia	HP:0000347	✓	✓	✓	✓
Broad thumbs	HP:0011304	✓	✗	✗	✓
Inheritance	X-linked recessive	✗	✗	✗	✗
Matching Accuracy		100.0%	66.7%	50.0%	83.3%
Precision (%)		100.0%	66.7%	50.0%	83.3%
Recall (%)		100.0%	66.7%	50.0%	83.3%

In Case 7, DeepSeek achieved a perfect score, accurately identifying all six phenotype terms from GeneBreaker’s reference, resulting in **100%** across matching accuracy, precision, and recall. **Doc2HPO** followed with strong performance at **83.3%**, while **ChatGPT-4** and **LLaMA 3.3** identified fewer correct terms, scoring **66.7%** and **50.0%** respectively. None of the tools correctly identified the **inheritance pattern**, which remains a common weakness across most cases.

Case 8:

“A 12-month-old male was seen for routine well child care. His gestation was notable for possible fetal hydrops, but this resolved spontaneously in the beginning of the 3rd trimester. Amniocentesis revealed a normal male karyotype. During the neonatal period, he was identified as having a heart murmur and unilateral cryptorchidism. An echocardiogram was performed and showed a patent ductus arteriosus (PDA) and mild pulmonary valve dysplasia. His developmental milestones were within the normal range, but were at the lower limits of normal for gross motor and language skills. On physical exam he was noted to have short stature with characteristic facial features including low-set, posteriorly rotated ears, widely-spaced eyes, a short and broad nose and wide neck”.

Term	GeneBreaker (Ref)	DeepSeek	ChatGPT-4	LLaMA 3.3	Doc2HPO
Short stature	HP:0004322	✓	✓	✓	✓
Congenital heart disease	HP:0032318	✗	✓	✗	✗
Wide neck	HP:0000475	✓	✓	✗	✓
Low set posteriorly rotated ears	HP:0000368	✗	✓	✗	✓
Widely spaced eyes	HP:0000316	✓	✓	✓	✓
Broad nose	HP:0000445	✓	✓	✗	✗
Short nose	HP:0003196	✗	✓	✗	✗
Inheritance	X-linked recessive	✗	✗	✗	✗
Matching Accuracy		71.4%	85.7%	42.9%	57.1%
Precision (%)		35.7%	46.2%	30.0%	50.0%
Recall (%)		71.4%	85.7%	42.9%	57.1%

In Case 8, **ChatGPT-4** demonstrated the strongest performance, correctly identifying 6 out of 7 phenotype terms with **85.7% matching accuracy and recall**, and **46.2% precision**. **DeepSeek** also performed well in recall but showed low precision. **Doc2HPO** offered a good balance between precision and recall (**50.0% and 57.1%** respectively). **LLaMA 3.3** identified fewer correct terms, trailing behind in all metrics. Once again, none of the tools successfully predicted the **inheritance pattern**, reflecting a broader limitation across models.

Case 9:

“A ten year old female was referred to Medical Genetics. Her clinical presentation included: Marfanoid habitus, a flat midface, severe myopia and a cleft palate. The family history revealed her father had mitral valve prolapse, sensorineural hearing loss, myopia and a high arched palate. A paternal aunt and the paternal grandfather had hearing loss and near sightedness”.

Term	GeneBreaker (Ref)	DeepSeek	ChatGPT-4o	LLaMA 3.3	Doc2HPO
Flat midface	HP:0011800	✓	✓	✓	✓
Sensorineural hearing loss	HP:0008625	✓	✓	✓	✓
Myopia	HP:0000545	✓	✓	✓	✓
Cleft palate	HP:0000175	✓	✓	✓	✓
Inheritance	X-linked recessive (hemizygous, de novo)	✗	✗	✗	✗
Matching Accuracy		100.0%	100.0%	100.0%	100.0%
Precision (%)		66.7%	66.7%	66.7%	66.7%
Recall (%)		100.0%	100.0%	100.0%	100.0%

In Case 9, all tools—**DeepSeek**, **ChatGPT-4o**, **LLaMA 3.3**, and **Doc2HPO**—correctly extracted **all four phenotype terms**, achieving a perfect **100% matching accuracy and recall**. However, none of the tools correctly identified the **inheritance pattern**, which led to a **precision score of 66.7%** across the board due to the extra, incorrect inheritance prediction. This case showcases excellent phenotype extraction performance while also emphasizing the ongoing challenge with inheritance interpretation.

Case 10:

“An 8 year old boy was referred to genetics and had the following clinical presentation: mild intellectual disability, short stature, microcephaly, triangular face, large and prominent ears, hypertelorism, long palpebral fissures and macrodontia. Both parents attended the appointment and the geneticist noted that the mother demonstrated similar craniofacial findings to the son. Upon further questioning and examination, it was revealed that the mother was similarly affected, displaying moderate intellectual disability, and overlap of symptoms with the child”.

Term	GeneBreaker (Ref)	DeepSeek	ChatGPT-4	LLaMA 3.3	Doc2HPO
Intellectual disability	HP:0001249	✓	✓	✓	✓
Short stature	HP:0004322	✓	✓	✓	✓
Microcephaly	HP:0000252	✓	✓	✓	✓
Triangular face	HP:0000325	✓	✓	✓	✓
Large and prominent ears	HP:0000400	✓	✓	✓	✓
Hypertelorism	HP:0000316	✓	✓	✓	✓
Long palpebral fissures	HP:0000637	✓	✓	✓	✓
Macrodontia	HP:0001572	✓	✓	✓	✓
Inheritance	Y-linked	✗	✗	✗	✗
Matching Accuracy		100.0%	100.0%	100.0%	100.0%
Precision (%)		100.0%	100.0%	100.0%	100.0%
Recall (%)		100.0%	100.0%	100.0%	100.0%

In Case 10, all tools—**DeepSeek**, **ChatGPT-4**, **LLaMA 3.3**, and **Doc2HPO**—correctly extracted all 8 HPO phenotype terms from the clinical case, achieving a perfect **100% accuracy, precision, and recall**. However, none of the tools correctly predicted the **Y-linked inheritance**, which is a consistent pattern across multiple cases. Despite this, their performance on phenotype recognition in this case is outstanding.

AI Assisted Re-Ranking of Variants – Llama 3.3 Case Results

Case 1:

1. ****Variant 1 (WAS)****: This variant is ranked as the most likely to cause the phenotypes due to its ACMG classification as 'Pathogenic' and its association with Wiskott-Aldrich syndrome, a condition that affects blood cells and the immune system. Although there are no literature counts for the specific phenotypes mentioned, the ClinGen gene-disease relationships indicate a definitive association with Wiskott-Aldrich syndrome, which can manifest with eczematoid dermatitis, thrombocytopenia, and abnormalities of the immune system. The pathogenic classification and the disease association make this variant a strong candidate.

2. **Variant 2 (PMS2)**: This variant is ranked second due to its association with Lynch syndrome and hereditary breast carcinoma, as indicated by ClinGen gene-disease relationships. The gene-phenotype literature counts show a significant number of papers related to neoplasms (2070 papers) and abnormalities of the immune system (9 papers), which aligns with some of the phenotypes mentioned. However, its ACMG classification as 'Uncertain significance' reduces its ranking compared to Variant 1.
3. **Variant 8 (SCN9A)**: This variant is ranked third because, despite its 'Uncertain significance' ACMG classification, it has a relatively high number of papers related to phenotypes such as short stature (4 papers), neoplasms (41 papers), and abnormalities of the immune system (2 papers). The presence of some relevant literature counts and its association with epilepsy (though refuted) suggest it could contribute to the phenotypes, especially considering the broader spectrum of conditions it might be involved in.
4. **Variant 7 (GPR101)**: Ranked fourth, this variant has an 'Uncertain significance' classification but shows associations with short stature (2 papers), neoplasms (48 papers), and abnormalities of the immune system (1 paper). The significant number of papers related to neoplasms is notable, but the lack of a definitive disease association in ClinGen reduces its ranking.
5. **Variant 9 (MXRA5)**: This variant is ranked fifth due to its 'Uncertain significance' classification and the presence of 20 papers related to neoplasms. However, it lacks specific associations with most of the phenotypes mentioned and does not have a definitive disease association in ClinGen.
6. **Variant 5 (ENOX2)**: Ranked sixth, this variant has a 'Likely benign' classification, which reduces its likelihood of causing the phenotypes. However, it has 20 papers related to neoplasms and 1 paper related to abnormalities of the immune system, suggesting some potential relevance.

7. **Variant 6 (CACNA1F)**: This variant is ranked seventh. With a 'Likely benign' classification and only 3 papers related to neoplasms and 1 paper related to abnormalities of the immune system, its potential impact on the phenotypes is considered lower.
8. **Variant 10 (SLU7)**: Ranked eighth, this variant has an 'Uncertain significance' classification but lacks significant literature counts for the phenotypes, with only 8 papers related to neoplasms.
9. **Variant 4 (POLR2J3)**: This variant is ranked ninth due to its 'Likely benign' classification and limited relevant literature counts, with only 3 papers related to neoplasms.
10. **Variant 3 (CNTNAP3)**: Ranked last, this variant has an 'Uncertain significance' classification but very limited relevant literature, with only 6 papers related to neoplasms and no other significant associations with the mentioned phenotypes.

Conclusion: Based on the analysis, **Variant 1 (WAS)** is the most likely to cause the phenotypes due to its pathogenic classification and association with Wiskott-Aldrich syndrome. Variants 2 (PMS2) and 8 (SCN9A) also show potential due to their literature counts and disease associations, despite their 'Uncertain significance' classifications. The ranking highlights the importance of considering both the ACMG classification and the gene-phenotype literature counts when assessing the potential of a variant to cause specific phenotypes. Further evaluation, including functional studies and family segregation analysis, would be necessary to confirm the pathogenicity of these variants in relation to the described phenotypes.

Case 2:

1. ****Variant 1 (MSH2)****: This variant is ranked as the most likely to cause the phenotypes due to its pathogenic classification by GeneBe and InterVar, and its association with a high number of neoplasms (4282 papers) and abdominal pain (15 papers). The presence of 2 papers related to abnormal eye morphology also suggests a possible link to this phenotype. The ClinGen gene-disease relationships confirm the association of MSH2 with Lynch syndrome and mismatch repair cancer syndrome, which can explain the high incidence of neoplasms. The gene-phenotype literature counts, such as the 224 papers related to clinical course, further support the potential impact of this variant on the overall clinical presentation.
2. **Variant 5 (MYH9)**: This variant is ranked second due to its likely benign classification, but its association with 3 papers related to abnormal eye morphology and 213 papers related to neoplasms. Although it is not considered pathogenic, the gene-phenotype literature counts suggest a possible link to certain phenotypes, such as abnormal eye morphology and neoplasms. The ClinGen gene-disease relationships confirm the

association of MYH9 with macrothrombocytopenia and granulocyte inclusions, which may contribute to the overall clinical presentation.

3. **Variant 4 (SCN9A):** This variant is ranked third due to its uncertain significance classification and its association with 1 paper related to abnormal eye morphology, 8 papers related to abdominal pain, and 41 papers related to neoplasms. The gene-phenotype literature counts suggest a possible link to certain phenotypes, such as abdominal pain and neoplasms. However, the ClinGen gene-disease relationships only refuted the association with epilepsy, which reduces the confidence in this variant's pathogenicity.
4. **Variant 9 (BACE2):** This variant is ranked fourth due to its uncertain significance classification and its association with 18 papers related to neoplasms. Although it does not have a clear link to other phenotypes, the gene-phenotype literature counts suggest a possible contribution to the development of neoplasms. However, the lack of other supporting evidence, such as ClinGen gene-disease relationships, reduces the confidence in this variant's pathogenicity.
5. **Variant 2 (POGLUT1):** This variant is ranked fifth due to its uncertain significance classification and its association with 1 paper related to abnormal eye morphology and 9 papers related to neoplasms. Although it has a possible link to abnormal eye morphology, the lack of other supporting evidence, such as ClinGen gene-disease relationships, reduces the confidence in this variant's pathogenicity.
6. **Variant 7 (FAM162A):** This variant is ranked sixth due to its uncertain significance classification and its association with 11 papers related to neoplasms. Although it has a possible link to neoplasms, the lack of other supporting evidence, such as ClinGen gene-disease relationships, reduces the confidence in this variant's pathogenicity.
7. **Variant 6 (TOP2B):** This variant is ranked seventh due to its likely benign classification and its association with 79 papers related to neoplasms. Although it has a possible link to neoplasms, the lack of other supporting evidence, such as ClinGen gene-disease relationships and gene-phenotype literature counts for other phenotypes, reduces the confidence in this variant's pathogenicity.
8. **Variant 10 (FARP2):** This variant is ranked eighth due to its likely benign classification and its association with 9 papers related to neoplasms. Although it has a possible link to neoplasms, the lack of other supporting evidence, such as ClinGen gene-disease relationships and gene-phenotype literature counts for other phenotypes, reduces the confidence in this variant's pathogenicity.
9. **Variant 8 (EXOSC7):** This variant is ranked ninth due to its likely benign classification and its association with only 1 paper related to neoplasms. The lack of other supporting

evidence, such as ClinGen gene-disease relationships and gene-phenotype literature counts for other phenotypes, reduces the confidence in this variant's pathogenicity.

10. **Variant 3 (HACL1):** This variant is ranked last due to its uncertain significance classification and its lack of association with any phenotypes, except for 2 papers related to neoplasms. The lack of other supporting evidence, such as ClinGen gene-disease relationships and gene-phenotype literature counts for other phenotypes, reduces the confidence in this variant's pathogenicity.

Conclusion: Based on the analysis, Variant 1 (MSH2) is the most likely to explain the phenotypes due to its pathogenic classification, association with a high number of neoplasms and abdominal pain, and supporting evidence from ClinGen gene-disease relationships and gene-phenotype literature counts. Variant 5 (MYH9) and Variant 4 (SCN9A) are also potential contributors to the phenotypes, but their likely benign or uncertain significance classifications reduce their confidence. The other variants have limited evidence supporting their pathogenicity, and therefore, are less likely to explain the phenotypes.

Case 3:

1. ****Variant 4 (MALT1)**:** This variant is ranked as the most likely to cause the phenotypes due to its ACMG classification as 'Pathogenic' and its association with combined immunodeficiency due to MALT1 deficiency, as indicated by ClinGen gene-disease relationships. The gene-phenotype literature counts also show a significant number of papers related to atopic dermatitis (10 papers), recurrent skin infections (7 papers), gastrointestinal inflammation (7 papers), and pneumonia (4 papers), suggesting a strong link between MALT1 and these phenotypes. The frameshift variant in MALT1 is likely to disrupt the function of the MALT1 protein, leading to immune system dysfunction and contributing to the development of these phenotypes.

2. **Variant 1 (PEPD):** This variant is ranked second due to its ACMG classification as 'Pathogenic' and its association with prolidase deficiency, a rare autosomal recessive disorder characterized by recurrent infections, skin lesions, and other symptoms. The gene-phenotype literature counts show a significant number of papers related to recurrent skin infections (16 papers), recurrent respiratory infections (7 papers), and pneumonia (2 papers), indicating a strong link between PEPD and these phenotypes. The missense variant in PEPD is likely to affect the function of the prolidase enzyme, leading to the development of these phenotypes.
3. **Variant 10 (CD247):** This variant is ranked third due to its ACMG classification as 'Likely pathogenic' and its association with immunodeficiency 25, as indicated by ClinGen gene-disease relationships. The gene-phenotype literature counts show a

significant number of papers related to pneumonia (5 papers) and gastrointestinal inflammation (1 paper), suggesting a possible link between CD247 and these phenotypes. The splice region variant in CD247 may affect the function of the CD247 protein, leading to immune system dysfunction and contributing to the development of these phenotypes.

4. **Variant 3 (HLA-B):** This variant is ranked fourth due to its ACMG classification as 'Likely benign', but its association with a large number of papers related to atopic dermatitis (12 papers), recurrent skin infections (11 papers), gastrointestinal inflammation (78 papers), and pneumonia (223 papers), as shown in the gene-phenotype literature counts. Although the variant is likely benign, the strong association between HLA-B and these phenotypes suggests that it may still play a role in the development of these conditions.
5. **Variant 2 (RYR1):** This variant is ranked fifth due to its ACMG classification as 'Uncertain significance' and its association with malignant hyperthermia and RYR1-related myopathy, as indicated by ClinGen gene-disease relationships. The gene-phenotype literature counts show a significant number of papers related to pneumonia (10 papers), but few papers related to the other phenotypes. The missense variant in RYR1 may affect the function of the RYR1 protein, but its relationship to the phenotypes is less clear.
6. **Variant 6 (SMPD1):** This variant is ranked sixth due to its ACMG classification as 'Likely benign' and its association with acid sphingomyelinase deficiency, as indicated by ClinGen gene-disease relationships. The gene-phenotype literature counts show a significant number of papers related to pneumonia (7 papers) and atopic dermatitis (3 papers), but few papers related to the other phenotypes. The 5' UTR premature start codon gain variant in SMPD1 may not significantly affect the function of the SMPD1 protein.
7. **Variant 5 (MYH6):** This variant is ranked seventh due to its ACMG classification as 'Likely benign' and its association with MYH-6 related congenital heart defects and dilated cardiomyopathy, as indicated by ClinGen gene-disease relationships. The gene-phenotype literature counts show few papers related to the phenotypes, suggesting a weak link between MYH6 and these conditions.
8. **Variant 7 (PKP1):** This variant is ranked eighth due to its ACMG classification as 'Benign' and its lack of association with any disease, as indicated by ClinGen gene-disease relationships. The gene-phenotype literature counts show few papers related to the phenotypes, suggesting a weak link between PKP1 and these conditions.
9. **Variant 8 (CADPS2):** This variant is ranked ninth due to its ACMG classification as 'Uncertain significance' and its lack of association with any disease, as indicated by

ClinGen gene-disease relationships. The gene-phenotype literature counts show no papers related to the phenotypes, suggesting no link between CADPS2 and these conditions.

10. **Variant 9 (SELENOM):** This variant is ranked tenth due to its ACMG classification as 'Uncertain significance' and its lack of association with any disease, as indicated by ClinGen gene-disease relationships. The gene-phenotype literature counts show only one paper related to pneumonia, suggesting a weak link between SELENOM and these conditions.

Conclusion: Based on the analysis of the variants, Variant 4 (MALT1) and Variant 1 (PEPD) are the most likely to cause the phenotypes 'Eczematoid dermatitis, Atopic dermatitis, Recurrent skin infections, Gastrointestinal inflammation, Recurrent respiratory infections, Pneumonia'. The strong association between MALT1 and PEPD and these phenotypes, combined with their ACMG classification as 'Pathogenic', suggests that they play a significant role in the development of these conditions. Variant 10 (CD247) and Variant 3 (HLA-B) may also contribute to the development of these phenotypes, although their relationship is less clear. The other variants are less likely to cause the phenotypes due to their benign or uncertain significance classification and lack of association with the phenotypes. Further studies are needed to confirm the role of these variants in the development of these conditions.

Case 4:

1. ****Variant 3 (CFTR)**:** This variant is ranked as the most likely to cause the phenotypes due to its pathogenic classification by GeneBe and strong evidence from the gene-phenotype literature counts. The CFTR gene is associated with multiple phenotypes, including short stature (10 papers), decreased body weight (92 papers), abnormality of the immune system (53 papers), pruritus (4 papers), and palmoplantar keratoderma (16 papers). The ACMG classification of this variant as 'Pathogenic' further supports its high ranking.
2. **Variant 4 (CFTR):** This variant is ranked second due to its pathogenic classification by GeneBe and InterVar, and its association with the CFTR gene, which has a high number of literature counts for relevant phenotypes. The missense variant has a strong ACMG classification, with multiple lines of evidence supporting its pathogenicity, including PM1, PM2, PP3_Strong, and PP5_Very_Strong. The CFTR gene's association with short stature, decreased body weight, abnormality of the immune system, pruritus, and palmoplantar keratoderma further supports this variant's high ranking.
3. **Variant 7 (SCN9A):** This variant is ranked third due to its uncertain significance classification by GeneBe and InterVar, but its association with the SCN9A gene, which has some literature counts for relevant phenotypes, such as short stature (4 papers), abnormality of the immune system (2 papers), and pruritus (9 papers). Although the

ACMG classification is not as strong as the top two variants, the gene-phenotype literature counts suggest a possible connection to the phenotypes.

4. **Variant 2 (PMS2):** This variant is ranked fourth due to its uncertain significance classification by GeneBe and InterVar, but its association with the PMS2 gene, which has some literature counts for relevant phenotypes, such as decreased body weight (8 papers) and abnormality of the immune system (9 papers). The ClinGen gene-disease relationships also suggest a possible connection to Lynch syndrome and hereditary breast carcinoma, although these are not directly related to the phenotypes of interest.
5. **Variant 9 (CACNA1F):** This variant is ranked fifth due to its likely benign classification by GeneBe and uncertain significance classification by InterVar. However, its association with the CACNA1F gene, which has a literature count for abnormality of the immune system (1 paper), and the ClinGen gene-disease relationship with CACNA1F-related retinopathy, suggests a possible connection to the phenotypes.
6. **Variant 1 (SASH1):** This variant is ranked sixth due to its benign classification by GeneBe and lack of strong ACMG classification. Although the SASH1 gene has some literature counts for relevant phenotypes, such as abnormality of the immune system (2 papers) and palmoplantar keratoderma (3 papers), the evidence is not as strong as the top variants.
7. **Variant 8 (ENOX2):** This variant is ranked seventh due to its likely benign classification by GeneBe and uncertain significance classification by InterVar. The ENOX2 gene has a literature count for abnormality of the immune system (1 paper), but the evidence is not as strong as the top variants.
8. **Variant 10 (MXRA5):** This variant is ranked eighth due to its uncertain significance classification by GeneBe and InterVar, but its association with the MXRA5 gene, which has a literature count for decreased body weight (1 paper). The lack of strong ACMG classification and limited gene-phenotype literature counts makes this variant less likely to cause the phenotypes.
9. **Variant 5 (CNTNAP3):** This variant is ranked ninth due to its uncertain significance classification by GeneBe and lack of strong ACMG classification. The CNTNAP3 gene has no literature counts for relevant phenotypes, making this variant unlikely to cause the phenotypes.
10. **Variant 6 (POLR2J3):** This variant is ranked tenth due to its likely benign classification by GeneBe and benign classification by InterVar. The POLR2J3 gene has no literature counts for relevant phenotypes, making this variant unlikely to cause the phenotypes.

Conclusion: Based on the ACMG classification, ClinGen gene-disease relationships, and gene-phenotype literature counts, Variants 3 and 4 (CFTR) are the most likely to cause the

phenotypes, followed by Variant 7 (SCN9A). The strong ACMG classification and high literature counts for relevant phenotypes associated with the CFTR gene make these variants the top candidates. Further analysis and functional studies are necessary to confirm the pathogenicity of these variants and their connection to the phenotypes.

Case 5:

1. ****Variant 1 (MECP2)**:** This variant is ranked as the most likely to cause the phenotypes due to its ACMG classification as 'Likely_pathogenic' and its association with a definitive disease relationship in ClinGen ('Rett syndrome'). The gene-phenotype literature counts also strongly support this ranking, with high counts for 'Developmental regression' (102 papers), 'Global developmental delay' (28 papers), 'Motor delay' (53 papers), 'Microcephaly' (93 papers), and 'Neurodevelopmental abnormality' (296 papers). The presence of a frameshift variant in the MECP2 gene, which is a well-established cause of Rett syndrome, further supports this ranking.
2. **Variant 2 (MAML1):** This variant is ranked second due to its ACMG classification as 'Pathogenic' and its association with 'Neoplasm of the genitourinary tract' (7 papers) in the gene-phenotype literature counts. Although MAML1 is not typically associated with neurodevelopmental disorders, the presence of a frameshift variant and the association with genitourinary tract neoplasms make it a potential contributor to the phenotypes.
3. **Variant 4 (ARL13A):** This variant is ranked third due to its ACMG classification as 'Pathogenic' and its potential association with ciliopathy disorders, which can cause developmental delays and neurodevelopmental abnormalities. Although there are no direct gene-phenotype literature counts for ARL13A, the presence of a splice acceptor variant and the potential for disrupted ciliary function make it a possible contributor to the phenotypes.
4. **Variant 9 (RYR1):** This variant is ranked fourth due to its ACMG classification as 'Uncertain_significance' and its association with 'Motor delay' (32 papers) and 'Neurodevelopmental abnormality' (3 papers) in the gene-phenotype literature counts. The presence of a missense variant in the RYR1 gene, which is associated with malignant hyperthermia and myopathy, makes it a potential contributor to the motor delay and neurodevelopmental abnormality phenotypes.
5. **Variant 8 (PKP2):** This variant is ranked fifth due to its ACMG classification as 'Uncertain_significance' and its association with 'Dysarthria' (1 paper) and 'Neoplasm of the genitourinary tract' (4 papers) in the gene-phenotype literature counts. Although PKP2 is primarily associated with cardiac disorders, the presence of a missense variant

and the potential for disrupted cellular adhesion make it a possible contributor to the phenotypes.

6. **Variant 7 (ELMO1):** This variant is ranked sixth due to its ACMG classification as 'Uncertain_significance' and its association with 'Developmental regression' (1 paper) and 'Neoplasm of the genitourinary tract' (2 papers) in the gene-phenotype literature counts. The presence of a missense variant in the ELMO1 gene makes it a potential contributor to the developmental regression and genitourinary tract neoplasm phenotypes.
7. **Variant 5 (OR8U1):** This variant is ranked seventh due to its ACMG classification as 'Uncertain_significance' and its association with 'Neoplasm of the genitourinary tract' (1 paper) in the gene-phenotype literature counts. Although OR8U1 is not typically associated with neurodevelopmental disorders, the presence of a frameshift variant makes it a possible contributor to the genitourinary tract neoplasm phenotype.
8. **Variant 10 (RYR1):** This variant is ranked eighth due to its ACMG classification as 'Likely_benign' and its association with 'Motor delay' (32 papers) and 'Neurodevelopmental abnormality' (3 papers) in the gene-phenotype literature counts. Although the RYR1 gene is associated with malignant hyperthermia and myopathy, the presence of a likely benign missense variant makes it less likely to contribute to the phenotypes.
9. **Variant 6 (PITPNM3):** This variant is ranked ninth due to its ACMG classification as 'Uncertain_significance' and the lack of gene-phenotype literature counts. The presence of a missense variant in the PITPNM3 gene makes it a possible contributor to the phenotypes, but the lack of supporting evidence makes it less likely.
10. **Variant 3 (FOXD4L5):** This variant is ranked tenth due to its ACMG classification as 'Likely_benign' and the lack of gene-phenotype literature counts. The presence of a frameshift variant in the FOXD4L5 gene makes it unlikely to contribute to the phenotypes.

Conclusion: Based on the analysis, Variant 1 (MECP2) is the most likely to cause the phenotypes due to its ACMG classification, ClinGen gene-disease relationships, and strong gene-phenotype literature evidence. Variants 2 (MAML1) and 4 (ARL13A) are also potential contributors due to their ACMG classifications and gene-phenotype literature counts. However, the lack of direct evidence and the complexity of the phenotypes make it challenging to determine a single causal variant. Further analysis and experimentation are necessary to confirm the contributions of these variants to the phenotypes.

Case 6:

1. ****Variant 1 (ABCD1)**:** This variant is classified as pathogenic by GeneBe and has a strong association with several phenotypes, including generalized hypotonia (3 papers), intellectual disability (26 papers), and autism (3 papers). The ClinGen gene-disease relationship for ABCD1 is definitive for adrenoleukodystrophy, which is a severe form of intellectual disability. Given the ACMG classification and the significant gene-phenotype literature counts, Variant 1 is the most likely to cause the observed phenotypes.

2. **Variant 8 (SCN9A):** This variant is classified as uncertain significance by both GeneBe and InterVar. However, it has a notable association with several phenotypes, including language impairment (4 papers), global developmental delay (3 papers), autism (5 papers), and EEG abnormality (4 papers). The gene-phenotype literature counts suggest a potential link between SCN9A and the observed phenotypes, particularly in terms of developmental delay and autism. Although the ACMG classification is not as strong as Variant 1, the literature evidence supports a possible association.
3. **Variant 9 (TTN):** This variant is classified as uncertain significance by both GeneBe and InterVar. It has associations with generalized hypotonia (7 papers), intellectual disability (2 papers), and autism (5 papers). The ClinGen gene-disease relationships for TTN include dilated cardiomyopathy and TTN-related myopathy, but there is limited evidence linking TTN to the observed phenotypes directly. However, the gene-phenotype literature counts suggest a potential link, particularly with regard to hypotonia and autism.
4. **Variant 3 (PMS2):** This variant is classified as uncertain significance by GeneBe and has a limited association with the observed phenotypes, including delayed speech and language development (1 paper), language impairment (1 paper), and autism (2 papers). The ClinGen gene-disease relationships for PMS2 include Lynch syndrome and hereditary breast carcinoma, which do not directly relate to the observed phenotypes. Given the limited gene-phenotype literature counts and the absence of a strong ACMG classification, Variant 3 is less likely to cause the observed phenotypes.
5. **Variant 4 (CNTNAP3):** This variant is classified as uncertain significance by GeneBe and has a limited association with the observed phenotypes, including delayed speech and language development (1 paper), language impairment (1 paper), and autism (2 papers). There is no ClinGen gene-disease relationship found for CNTNAP3, and the gene-phenotype literature counts are limited. Therefore, Variant 4 is less likely to cause the observed phenotypes.
6. **Variant 6 (CACNA1F):** This variant is classified as likely benign by GeneBe and has a limited association with the observed phenotypes, including global developmental delay (1 paper) and autism (3 papers). The ClinGen gene-disease relationship for CACNA1F is definitive for CACNA1F-related retinopathy, which does not directly relate to the

observed phenotypes. Given the limited gene-phenotype literature counts and the ACMG classification as likely benign, Variant 6 is unlikely to cause the observed phenotypes.

7. **Variant 5 (POLR2J3):** This variant is classified as likely benign by both GeneBe and InterVar, and there is no association with the observed phenotypes in the gene-phenotype literature. The ClinGen gene-disease relationships do not include any relevant diseases. Therefore, Variant 5 is unlikely to cause the observed phenotypes.
8. **Variant 7 (ENOX2):** This variant is classified as likely benign by GeneBe and has no association with the observed phenotypes in the gene-phenotype literature. There is no ClinGen gene-disease relationship found for ENOX2. Given the lack of evidence, Variant 7 is unlikely to cause the observed phenotypes.
9. **Variant 10 (TTN):** This variant is classified as likely benign by both GeneBe and InterVar, and although it has some association with the observed phenotypes, such as generalized hypotonia (7 papers) and autism (5 papers), the ACMG classification and the presence of another variant (Variant 9) in the same gene with a higher likelihood of pathogenicity make Variant 10 less likely to be the causative variant.
10. **Variant 2 (ABCD1):** This variant is identical to Variant 1 and has the same evidence supporting its pathogenicity. However, since it is a duplicate, it does not add new information to the analysis.

Conclusion: Based on the ACMG classification, ClinGen gene-disease relationships, and gene-phenotype literature counts, Variant 1 (ABCD1) is the most likely to cause the observed phenotypes, including delayed speech and language development, generalized hypotonia, intellectual disability, and autism. Variants 8 (SCN9A) and 9 (TTN) also show potential associations with the observed phenotypes, particularly in terms of developmental delay and autism. However, the evidence is not as strong as for Variant 1. The other variants have limited or no evidence supporting their association with the observed phenotypes. Therefore, further investigation of Variant 1 (ABCD1) is warranted to confirm its role in causing the observed phenotypes.

Case 7:

1. ****Variant 1 (EP300)**:** This variant is ranked as the most likely to cause the phenotypes due to its ACMG classification as 'Pathogenic' and its significant gene-phenotype literature evidence. The literature counts show a strong association between EP300 and the phenotypes, with 9 papers linking EP300 to 'Intellectual disability, moderate', 1 paper to 'Abnormal facial shape', 1 paper to 'Downslanted palpebral fissures', 1 paper to 'Prominent nasal bridge', 3 papers to 'Micrognathia', 40 papers to 'Broad thumb', and 1 paper to 'Absent thumb'. Additionally, ClinGen

gene-disease relationships indicate a 'Definitive' relationship between EP300 and Rubinstein-Taybi syndrome, which is characterized by many of the listed phenotypes.

2. **Variant 2 (CADPS2):** Although this variant has an ACMG classification of 'Uncertain significance', it is still considered due to its association with 'Intellectual disability, moderate' as shown by 1 paper in the gene-phenotype literature count. However, the lack of literature evidence for other phenotypes and the absence of a ClinGen gene-disease relationship makes it less likely compared to Variant 1.
3. **Variant 6 (MEP1B):** With an ACMG classification of 'Likely pathogenic', this variant could potentially contribute to the phenotypes. However, the gene-phenotype literature counts do not show any direct links between MEP1B and the listed phenotypes, which lowers its ranking compared to Variant 1.
4. **Variant 8 (ACAD10):** This variant has an ACMG classification of 'Uncertain significance' and no gene-phenotype literature evidence supporting its link to the listed phenotypes. The lack of strong evidence places it below Variants 1, 2, and 6 in the ranking.
5. **Variant 10 (ITGAL):** Similar to Variant 8, this variant has an ACMG classification of 'Uncertain significance' but lacks gene-phenotype literature evidence, making it less likely to cause the phenotypes.
6. **Variant 4 (TRIB3):** With an ACMG classification of 'Uncertain significance' and no supporting gene-phenotype literature evidence, this variant is ranked lower.
7. **Variant 5 (KIF27):** This variant also has an ACMG classification of 'Uncertain significance' and lacks direct links to the phenotypes in the gene-phenotype literature counts, placing it in a lower rank.
8. **Variant 9 (SLC2A5):** The 'Uncertain significance' ACMG classification and absence of gene-phenotype literature evidence make this variant less likely to be the cause.
9. **Variant 3 (SMC4):** With an ACMG classification of 'Likely benign', this variant is considered less likely to contribute to the phenotypes, especially given the lack of supporting literature evidence.
10. **Variant 7 (LRP6):** This variant, classified as 'Likely benign', is the least likely due to its ACMG classification and the absence of gene-phenotype literature evidence linking it to the phenotypes.

In conclusion, Variant 1 in the EP300 gene is the most likely to cause the observed phenotypes due to its pathogenic classification and strong support from gene-phenotype literature evidence. While other variants have uncertain classifications, they lack the robust literature support seen with EP300, making Variant 1 the primary candidate for explaining the phenotypes 'Intellectual disability, moderate, Abnormal facial shape, Photophobia, Downslanted palpebral fissures,

Prominent nasal bridge, Micrognathia, Broad thumb, Absent thumb, Thick lower lip vermillion'. Further investigation into EP300 and its relationship to these phenotypes is warranted.

Case 8:

1. ****Variant 2 (PMS2)****: This variant is ranked as the most likely to cause the phenotypes due to its ACMG classification as 'Uncertain significance' with a 'PVS1_Moderate' supporting criterion, which suggests a possible pathogenic role. Additionally, the gene-phenotype literature counts show associations with several phenotypes, including Cryptorchidism (1 paper), Rhabdomyosarcoma (7 papers), Global developmental delay (2 papers), and Low-set ears (1 paper). The ClinGen gene-disease relationships also establish PMS2 as a definitive gene for Lynch syndrome and mismatch repair cancer syndrome 1, indicating its potential to contribute to developmental and cancer-related phenotypes.

2. **Variant 4 (HTT)**: This variant is ranked second due to its ACMG classification as 'Uncertain significance' with a 'PVS1_Strong' supporting criterion, indicating a possible frameshift variant. The gene-phenotype literature counts show associations with Global developmental delay (3 papers), Delayed speech and language development (5 papers), Short stature (2 papers), and Short neck (1 paper). Although HTT is primarily associated with Huntington disease, the variant's potential pathogenicity and the gene's involvement in developmental delay and speech-language development make it a plausible candidate.
3. **Variant 1 (CACNA1F)**: This variant is ranked third, with an ACMG classification as 'Likely benign' but still considered due to its association with Strabismus (16 papers) and Congenital blindness (118 papers) in the gene-phenotype literature counts. However, the lack of strong ACMG evidence and limited associations with other phenotypes reduce its ranking.
4. **Variant 10 (AHI1)**: This variant is ranked fourth, with an ACMG classification as 'Likely benign' but associations with Strabismus (4 papers), Low-set ears (1 paper), Short neck (1 paper), and Congenital blindness (1 paper) in the gene-phenotype literature counts. AHI1's established role in Joubert syndrome 3 (a definitive ClinGen gene-disease relationship) contributes to its consideration, despite the relatively weaker ACMG classification.
5. **Variant 3 (CNTNAP3)**: This variant is ranked fifth, with an ACMG classification as 'Uncertain significance' but limited gene-phenotype literature associations, primarily with Delayed speech and language development (1 paper). The lack of strong evidence reduces its ranking.
6. **Variant 6 (ENOX2)**: This variant is ranked sixth, with an ACMG classification as 'Likely benign' and a limited association with Rhabdomyosarcoma (1 paper). The evidence is too weak to strongly support its involvement in the phenotypes.

7. **Variant 7 (PODN):** This variant is ranked seventh, with an ACMG classification as 'Uncertain significance' but no significant gene-phenotype literature associations. The evidence is insufficient to support its involvement in the phenotypes.
8. **Variant 8 (PODN):** This variant is ranked eighth, with an ACMG classification as 'Likely benign' and no significant gene-phenotype literature associations, making it an unlikely candidate.
9. **Variant 9 (PODN):** This variant is ranked ninth, with an ACMG classification as 'Benign' and no significant gene-phenotype literature associations, providing no evidence for its involvement in the phenotypes.
10. **Variant 5 (POLR2J3):** This variant is ranked last, with an ACMG classification as 'Likely benign' and no significant gene-phenotype literature associations, offering no support for its involvement in the phenotypes.

Conclusion: Based on the comprehensive analysis, **Variant 2 (PMS2)** is the most likely to cause the phenotypes due to its potential pathogenicity, associations with multiple phenotypes, and the gene's established role in cancer syndromes. **Variant 4 (HTT)** is also a plausible candidate, considering its potential frameshift variant and associations with developmental delay and speech-language development. While **Variant 1 (CACNA1F)** and **Variant 10 (AHI1)** show some promise due to their gene-phenotype literature associations, the evidence is not as strong as for Variants 2 and 4. The remaining variants are less likely to contribute to the phenotypes, primarily due to weaker ACMG classifications and limited gene-phenotype literature associations. Further investigation and functional studies are necessary to confirm the roles of these variants in the observed phenotypes.

Case 9:

1. ****Variant 1 (COL2A1)**:** This variant is ranked as the most likely to cause the phenotypes due to its ACMG classification as 'Pathogenic' and its association with multiple phenotypes, including 'Polyhydramnios', 'Abnormal facial shape', 'Myopia', 'Cleft palate', 'Astigmatism', 'Hearing impairment', 'Visual impairment', 'Sensorineural hearing impairment', and 'Strabismus'. The gene-phenotype literature counts show a significant number of papers associated with these phenotypes, especially 'Myopia' (98 papers), 'Cleft palate' (78 papers), and 'Hearing impairment' (137 papers). ClinGen gene-disease relationships also indicate a definitive association with several diseases, including spondyloepiphyseal dysplasia congenita and Stickler syndrome type 1.
2. **Variant 8 (PHKA2):** This variant is ranked second due to its association with 'Hearing impairment' (3 papers) and 'Sensorineural hearing impairment' (2 papers), although its ACMG classification is 'Likely benign'. However, the InterVar result is 'Uncertain significance', indicating some potential pathogenicity. The lack of association with other

phenotypes and the relatively low gene-phenotype literature counts make it less likely to be the primary cause of the overall phenotypic profile.

3. **Variant 4 (TRIB3)**: This variant is ranked third due to its association with 'Hearing impairment' (1 paper) and 'Visual impairment' (3 papers), but its ACMG classification is 'Uncertain significance'. The lack of association with other phenotypes and the relatively low gene-phenotype literature counts make it less likely to be the primary cause of the overall phenotypic profile.
4. **Variant 5 (KIF27)**: This variant is ranked fourth due to its association with 'Cleft palate' (1 paper), but its ACMG classification is 'Uncertain significance'. The lack of association with other phenotypes and the relatively low gene-phenotype literature counts make it less likely to be the primary cause of the overall phenotypic profile.
5. **Variant 7 (RGPD4)**: This variant is ranked fifth due to its association with 'Cleft palate' (1 paper), but its ACMG classification is 'Uncertain significance'. The lack of association with other phenotypes and the relatively low gene-phenotype literature counts make it less likely to be the primary cause of the overall phenotypic profile.
6. **Variant 10 (MEP1B)**: This variant is ranked sixth due to its ACMG classification as 'Likely pathogenic', but it lacks association with any of the phenotypes. The gene-phenotype literature counts show no papers associated with any of the phenotypes.
7. **Variant 2 (PRAMEF10)**: This variant is ranked seventh due to its ACMG classification as 'Uncertain significance' and the lack of association with any of the phenotypes. The gene-phenotype literature counts show no papers associated with any of the phenotypes.
8. **Variant 3 (SMC4)**: This variant is ranked eighth due to its ACMG classification as 'Likely benign' and the lack of association with any of the phenotypes. The gene-phenotype literature counts show no papers associated with any of the phenotypes.
9. **Variant 6 (CADPS2)**: This variant is ranked ninth due to its ACMG classification as 'Uncertain significance' and the lack of association with any of the phenotypes. The gene-phenotype literature counts show no papers associated with any of the phenotypes.
10. **Variant 9 (PHKA2)**: This variant is ranked tenth due to its identical characteristics with Variant 8, but its lower ranking is due to the presence of another variant (Variant 8) with the same gene and similar characteristics.

In conclusion, Variant 1 (COL2A1) is the most likely to cause the phenotypes 'Polyhydramnios, Abnormal facial shape, Myopia, Cleft palate, Astigmatism, Mitral regurgitation, Hearing impairment, Visual impairment, Sensorineural hearing impairment, Strabismus' due to its pathogenic ACMG classification, significant gene-phenotype literature counts, and association with multiple phenotypes. While other variants show some association with individual

phenotypes, they lack the comprehensive profile seen with Variant 1. Therefore, COL2A1 is the primary candidate gene for explaining the overall phenotypic profile.

Case 10:

1. ****Variant 1 (ANKRD11)**:** This variant is ranked as the most likely to cause the phenotypes due to its ACMG classification as 'Pathogenic' and its association with 'KBG syndrome' through ClinGen gene-disease relationships. The gene-phenotype literature counts also show strong evidence for ANKRD11's involvement in intellectual disability, mild (21 papers), short stature (61 papers), microcephaly (4 papers), hypertelorism (5 papers), and long face (5 papers). The presence of a frameshift variant in ANKRD11, which is predicted to be pathogenic, further supports its likelihood of causing these phenotypes.

2. **Variant 8 (TTN):** Although not directly associated with all the listed phenotypes, Variant 8 is ranked second due to its uncertain significance classification and the presence of some gene-phenotype literature counts, such as 2 papers for intellectual disability, mild, and 2 papers for otitis media. TTN is known for its involvement in various muscular and cardiac disorders, but its potential impact on the listed phenotypes cannot be entirely ruled out, given the existing, albeit limited, literature evidence.
3. **Variant 3 (ROR2):** This variant is likely benign according to GeneBe and has an uncertain significance classification by InterVar. However, ROR2 has some gene-phenotype literature counts, particularly for short stature (38 papers) and hypertelorism (6 papers), which relate to the given phenotypes. The missense variant in ROR2 could potentially contribute to the phenotypes, but its likelihood is lower than that of Variant 1 due to its less severe classification and fewer relevant literature counts.
4. **Variant 4 (SH3BP2):** Ranked fourth, Variant 4 is considered benign by GeneBe and has uncertain significance by InterVar. SH3BP2 has some relevant gene-phenotype literature, including 6 papers for abnormal maxilla morphology and 2 papers for long face, as well as 1 paper for widely spaced teeth. Although the evidence is less strong compared to Variant 1, SH3BP2's association with certain phenotypes, like abnormal maxilla morphology, keeps it in consideration.
5. **Variant 5 (CACNA1F):** This variant is ranked fifth, with a likely benign classification and uncertain significance by InterVar. CACNA1F is associated with a definitive gene-disease relationship for 'CACNA1F-related retinopathy' and has 3 papers related to hyperopic astigmatism. However, its connection to the broader set of phenotypes is less direct and robust than the variants ranked above it.
6. **Variant 6 (MUC7):** With an uncertain significance classification and a stop gained variant, MUC7 is ranked sixth. It has some relevant literature for otitis media (3 papers)

and long face (1 paper). However, its overall connection to the listed phenotypes is weaker compared to the higher-ranked variants.

7. **Variant 2 (ABL1)**: This variant is classified as uncertain significance by both GeneBe and InterVar. ABL1 has limited but some relevant gene-phenotype literature counts, such as 2 papers for short stature and 1 paper for microcephaly. Its connection to the broader phenotypic profile is less pronounced, hence its seventh ranking.
8. **Variant 9 (ERVW-1)**: Ranked eighth, Variant 9 has an uncertain significance classification and a stop gained variant. ERVW-1 lacks direct and strong gene-phenotype literature connections to the given phenotypes, making it a less likely cause.
9. **Variant 10 (ENOX2)**: This variant is ranked ninth, with a likely benign classification and uncertain significance by InterVar. ENOX2 does not have significant gene-phenotype literature counts that directly relate to the given phenotypes, making it less likely to be the causal variant.
10. **Variant 7 (ENTPD8)**: Lastly, Variant 7, with an uncertain significance classification and frameshift variant, lacks significant gene-phenotype literature connections to the listed phenotypes. Thus, it is considered the least likely to cause the observed phenotypes.

Conclusion: Based on the ACMG classification, ClinGen gene-disease relationships, and the gene-phenotype literature counts, **Variant 1 (ANKRD11)** is the most likely to explain the phenotypes of intellectual disability, mild, short stature, microcephaly, abnormal maxilla morphology, otitis media, hypertelorism, upslanted palpebral fissure, hyperopic astigmatism, long face, and widely spaced teeth. The strong pathogenic classification and extensive literature evidence supporting ANKRD11's involvement in many of these phenotypes make it the primary candidate for the observed conditions. Further genetic and clinical evaluation would be necessary to confirm the role of ANKRD11 and potentially other variants in the etiology of these phenotypes.